MRI BRAIN CONT SPECTRO

Patient Id: GENRAD-SUB-681417	**Study Date:** 10 Oct 2026, 11:05
Patient Name: DALBEERO 74Y F	**Referring Doctor:**
Age / Sex: 74/Female	**Investigation:** MRI BRAIN CONT SPECTRO

Technique

Multiplanar, multisequential MRI of the brain was performed before and after intravenous contrast administration, along with MR spectroscopy.

REPORT

Findings

A relatively well-defined intra-axial lesion is noted in the right parietal parasagittal region, measuring approximately 19 × 18 × 28 mm (AP × CC × transverse). The lesion appears hypointense on T1-weighted images and hyperintense on T2-weighted images, with partial suppression on FLAIR images. It demonstrates restricted diffusion on DWI and peripheral enhancement on post-contrast images.

Significant surrounding perilesional white matter edema is noted, causing mild mass effect with approximately 3.5 mm midline shift towards the left.

MR spectroscopy demonstrates an NAA peak along with a lipid-lactate peak. The spectroscopic findings are inconclusive and do not permit definitive characterization of the lesion.

Associated signal changes are noted in the overlying calvarial bone, which appears abnormal on T2-weighted images and demonstrates mild post-contrast enhancement, with associated osteolytic changes. These findings may represent postoperative changes or direct involvement by an underlying neoplastic process. Clinical and surgical history correlation is advised.

The remaining cerebral parenchyma appears unremarkable.

The bilateral gangliothalamic complexes are normal.

The ventricular system is normal in size and configuration.

The remaining midline structures are unremarkable.

The cerebellum and posterior fossa structures appear normal.

Impression

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1. Right parietal parasagittal intra-axial lesion measuring approximately 19 × 18 × 28 mm, demonstrating restricted diffusion and peripheral contrast enhancement, with significant surrounding white matter edema and mild mass effect, causing approximately 3.5 mm midline shift towards the left.

2. Differential considerations include an infective lesion (particularly an abscess) and a neoplastic etiology, including glioma. The available MR spectroscopy findings are inconclusive and do not reliably differentiate between these possibilities.

3. Associated osteolytic changes and abnormal signal with mild enhancement in the overlying calvarium, possibly representing postoperative changes or neoplastic involvement.

ADVICE

Repeat MR spectroscopy with multivoxel acquisition and appropriate clinical, laboratory and imaging correlation. Histopathological evaluation, with microbiological assessment if infection is suspected, is recommended for definitive diagnosis.

Signature



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